

Nizar Esmat Fawzy

Software Engineer
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PROFESSIONAL SUMMARY

Software Engineer with 2 years of experience building and shipping production systems. Owns API design, PostgreSQL schema architecture, and integrations across 50+ REST endpoints and 5 client platforms in a 10+ engineer team. Strong CS fundamentals – data structures, algorithms, OOP, SOLID, system design – sharpened by ECPC competitive programming, plus TypeScript, Node.js, NestJS, Python/Django, and Socket.io.

EDUCATION

Nahda University Sep 2019 – Jul 2023
Bachelor of Computer Science Beni-Suef, Egypt

Information Technology Institute (ITI) Nov 2024 – Mar 2025
Full Stack Development Track (Python) Smart Village, Egypt

- Completed a competitive full-stack training program: designed a normalized PostgreSQL schema for 10k+ users and built RESTful APIs in Python/Django with ETL-style data workflows, contributing to architecture and data modeling decisions.

Certifications: Python Professional Track (ITI) | SQL & PostgreSQL (Udemy) | Node.js (Udemy) | MERN Stack (Route Academy)

PROFESSIONAL EXPERIENCE

Opream May 2025 – Present
Software Engineer Cairo, Egypt

- Design and ship 50+ RESTful API endpoints across 5 production platforms in TypeScript, NestJS, PostgreSQL, and React, owning each feature from API contract and data model design through testing and release.
- Architected **Mostaqbel Watan**, a private social platform serving 1,000+ active users, with Socket.io private/group messaging, Jitsi video meetings, and an automated Facebook Graph API pipeline ingesting group posts.
- Built **Onway** (ride-sharing with Fawry payments) and **Al-Azhar & Bingo** (lesson/library management); on **Event Ticketing**, eliminated inventory lock-up with a scheduled job releasing stale “pending” reservations every 10 minutes.
- Improve query latency by normalizing schemas, adding targeted indexes, and eliminating N+1 patterns; cover critical paths with Jest unit and integration tests and review teammates’ pull requests to catch defects before release.

iSchool Jul 2024 – Nov 2024
Software Instructor (Part-time) Cairo, Egypt

- Taught programming fundamentals and algorithmic problem solving, breaking down core CS concepts for students.

WE School Aug 2023 – Jun 2024
Web Development Instructor Cairo, Egypt

- Taught web development with HTML, CSS, and JavaScript, guiding students from fundamentals to working projects.

PROJECTS

ECPC Movement Control | *NestJS, PostgreSQL, TypeORM, Socket.io, React, JWT* GitHub

- Built a real-time volunteer dispatch system for the Egyptian Collegiate Programming Contest: a scoring algorithm auto-assigns requests by hall proximity, gender match, and idle time, backed by a versioned TypeORM/PostgreSQL schema, an audit trail, and live Socket.io updates under JWT-secured roles.

Chat Application | *React, Node.js, Express.js, Socket.io, MongoDB, Cloudinary* GitHub Postman

- Engineered a full-stack real-time chat application (React/Zustand, Node/Express/Socket.io) with private messaging, group channels, and live presence; hardened the API with HTTP-only cookie JWT auth, Arcjet rate limiting, and Cloudinary-backed media storage with full CRUD.

VOLUNTEERING & ACHIEVEMENTS

2nd Place – Nahda University ECPC Qualifiers (2021, 2022); **Organizing Volunteer** – ICPC/ECPC/ACPC, 2022–2024.

300+ algorithmic problems solved in C++ on Codeforces – data structures, algorithms, and complexity analysis.

TECHNICAL SKILLS

Programming Languages: TypeScript, JavaScript, Python, SQL, C++
Backend: Node.js, NestJS, Express.js, Django, REST APIs, Socket.io, TypeORM, JWT Authentication
Frontend: React, HTML, CSS
Databases: PostgreSQL, MongoDB, Data Modeling, Indexing, Query Optimization
Computer Science: Data Structures & Algorithms, Problem Solving, OOP, SOLID, Design Patterns
Testing & DevOps: Jest, Unit & Integration Testing, Git, GitHub, Docker, CI/CD, Linux

LANGUAGES

Arabic (Native) | English (Fluent)